NARRATIVE QUARTERLY REPORT (OCTOBER–DECEMBER) 2025

Project — In situ monitoring of water quality in Cuban bays: creating and promoting the use of scientific tools

Participating institutions

- University of Havana (UH)
- Université Libre de Bruxelles (Belgium)

Cooperating institutions

- Center for Environmental Research and Transport Management (CIMAB)
- ESPOLETA Tecnologías (SME)

Other collaborating institutions

- José Antonio Echeverría Technological University of Havana (Cujae)
- Havana Bay State Management Group
- Université Catholique de Louvain

Project coordinators

Cuban side: Dr. C. Ernesto Altshuler Álvarez
Foreign side: Dr. Denis Terwagne

Participants — Dr. Geydis Fundora Nevot, Dr. Florence Degavre, Dr. Stamatios Nicolis, Dr. Alexandre Campo, Lic. Daniela Martínez Ortiz (PhD candidate), Eng. Alex Rivera Rivera (PhD candidate), MSc. Luis Abel Rodríguez de Torner (PhD candidate), Dr. Aramis Rivera Denis, Eng. Orestes Chávez Linares, Dr. Dayaris Hernández.

Duration — Five years

General objective — To provide the Cuban scientific-social ecosystem with a set of tools that enables in situ monitoring of water-quality parameters in Cuban bays, in a sustainable way and connected to local communities.

Specific objectives

- Design and build a wave channel at the Faculty of Physics, UH
- Design and build a buoy prototype
- Set up a server with a website to receive data from the buoys and from the community
- Establish a methodology for using the buoys
- Train the corresponding human resources

Fulfillment of the quarterly objectives

Objective 1. Drafting the second-year purchase list, customs procedures, and receiving equipment

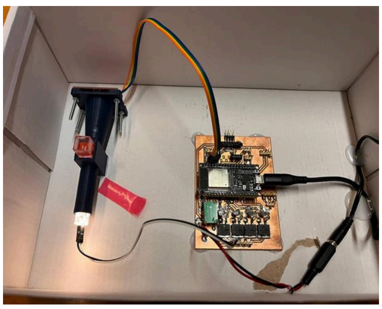
Import authorization had been obtained, on the Cuban side, for the first container from the Université Libre de Bruxelles. However, formalities on the Belgian side have also taken too long, so we now expect the container to arrive within the first four months of 2026. The first shipment from Panama was successfully cleared from the warehouses in Guanabacoa. It included four electric mopeds, a generator, and one spectrophotometer; the latter was installed at the Zeolite Engineering Laboratory (IMRE), under the supervision of Dr. Aramis Rivera and Dr. Dayaris Hernández, along with E. Altshuler. During the last quarter, constant negotiation continued with suppliers of specialized equipment for the second container from Belgium and the second shipment from Panama. After multiple procedures (including contracting an alarm-installation service), the alarm box was successfully cleared from Customs in early September 2025. Installation began, carried out by the project's own members with the help of students, but it has not yet been possible to complete the job due to technical problems. Finally, it should be clarified that E. Altshuler personally carried out the difficult and lengthy procedures needed to register the project's electric tricycle in his own name — only the first step in a long process for the tricycle to eventually be registered in the name of the University of Havana, for which we are requesting all possible support from the University.

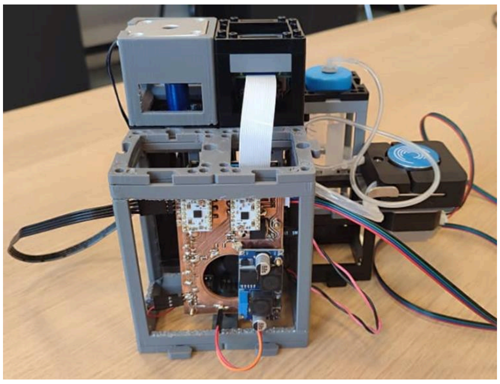
Objective 2. Detecting and addressing vulnerabilities in the Physics building that endanger the project

It should be noted that efforts have been under way to eliminate the vulnerabilities of the Physics building on the University of Havana's hill campus, where multiple windows in very poor condition due to termite damage make it easy for intruders to get in. This is a VERY SERIOUS situation. Action began with a letter to the Dean of Physics, who in turn raised the issue with the University's Rector. If these vulnerabilities (especially the ground-floor windows) are not resolved in time, the physical safety of the material investment associated with the Belgian project is at very serious risk. This is a matter of the UTMOST IMPORTANCE and URGENCY, one that could call into question our ability to carry the ARES project forward.

Objective 3. Continued instrumentation for in situ measurements

Throughout the period, multiple tests were run on the buoys' LoRa-based communication system, involving PhD candidates Alex Rivera, Daniela Martínez, and Luis Abel Rodríguez, along with Computer Science student Raciel Pupo. In particular, the resilience of the communication system with 12-volt batteries was confirmed on land for short-duration blackouts. However, they are not able to withstand the multiple, longer-duration blackouts that have been occurring toward the end of 2025, so the system was out of service by the end of 2025 and remains out of service. Especially during their doctoral stay at the Université Libre de Bruxelles, PhD candidates Alex Rivera and Daniela Martínez made notable progress on two key fronts (see figure below): (a) design and construction of several portable spectrophotometer prototypes for in situ measurements of marine water quality; (b) design and construction of a portable microscope prototype incorporating artificial intelligence, for in situ measurements of marine water quality. Additionally, Physics graduate Marlon Mederos continued tests, both in the lab and in Havana Bay waters, on the possibility of using LoRa signal-reception strength as a wireless measure of water conductivity/salinity. Tests continue to advance, still without definitive or publishable results.





Prototypes of new equipment for in situ measurements of marine water quality, built by PhD candidates Alex Rivera and Daniela Martínez-Ortiz. Left: one of the low-cost spectrophotometer prototypes. Right: low-cost microscope prototype incorporating artificial intelligence.

Objective 4. Academic exercises carried out by the PhD candidates involved in the Project

PhD candidate Daniela Martínez successfully passed the digital-fabrication course at the Université Libre de Bruxelles, and presented her final project (the smart microscope). In addition, all PhD candidates took part in writing a review article on in situ measurement of marine water quality, to be submitted for publication in the first half of 2026.

Objective 5. Preliminary studies on the effect of copper on marine corrosion

With the collaboration of PhD candidate Luis Abel Rodríguez, preliminary trials were carried out in Havana Bay suggesting that a copper mesh can substantially stop biological colonization (biofouling) of sensors in the marine environment, at roughly one meter below the surface.

Objective 6. Continued research in the social sciences

During the period under review, progress was made on the doctoral thesis objective of PhD candidate Mayelín García (CIMAB), related to identifying the needs and expectations of social actors in terms of knowledge, communication, and participation, related to water quality in Havana Bay. Community surveys continued, along with contacts with digital-fabrication projects such as "Espacios Creativos," which links up with our own project. Dr. Geydis Fundora and PhD candidate Mayelín García took part in these activities.

Sincerely,
Dr. C. Ernesto Altshuler
Project Director, Cuban side.

English translation of the original Spanish narrative report submitted to ARES, prepared for the CubanWaterLab website.